

Analytical & Methodological Audit Dossier

Exact mathematical proofs, variance identities, direct standardization formulas, and automated unit test code traceability.

1. Mathematical Proofs & Variance Identities

Proof 1.1: Exact Three-Factor Efficiency Decomposition

Cost per Approved Conversion decomposes as: $\text{Cost/Approved} = (\text{CPM} / 1000) \times (1 / \text{CTR}) \times (1 / \text{ConvRate})$. Under sequential substitution attribution from Campaign 936 (\$15.811) to Campaign 1178 (\$63.833):

- **Lever 1 (CPM Effect)** = \$12.072 - \$15.811 = **-\$3.739**
- **Lever 2 (CTR Effect)** = \$16.732 - \$12.072 = **+\$4.660**
- **Lever 3 (Conv Effect)** = \$63.833 - \$16.732 = **+\$47.101**
- **Total Explained Variance** = -\$3.739 + \$4.660 + \$47.101 = **+\$48.022**
- **Reconciliation Residual Error** = $|\Delta C - (\sum \text{Levers})|$ = **\$0.00000000**.

Proof 1.2: Direct Standardization & Confounding Adjustment

Weights: $w_i = \text{Clicks}_i / \sum \text{Clicks}$. Adjusted Cost = $\sum (w_i \times \text{Cost}_i)$.

- **Campaign 936 Adjusted:** \$23.69 (vs. \$15.81 raw)
- **Campaign 1178 Adjusted:** \$78.16 (vs. \$63.83 raw)
- **Ratio:** 3.30x unit cost penalty persists under identical demographic weighting.

Proof 1.3: Two-Proportion Hypothesis Testing & Normal Approximation

Pooled conversion rate: $p_{\text{hat}} = (x_1 + x_2) / (n_1 + n_2)$. Standard error: $SE_{\text{pool}} = \sqrt{p_{\text{hat}}(1-p_{\text{hat}})(1/n_1 + 1/n_2)}$. Test statistic $z = (p_1 - p_2) / SE_{\text{pool}}$. Two-tailed p-value = $2 \times (1 - \Phi(|z|))$. Confidence Interval 95% = $(p_1 - p_2) \pm 1.96 \times SE_{\text{diff}}$.

2. Code, Metric & File Traceability Matrix

Analytical Dimension	Python Module	SQL Pipeline	Export Target	Verification Test
Ratio-of-Sums Scorecard	src/metrics.py	sql/03_metrics.sql	exports/campaign_scorecard.csv	tests/test_metrics.py (PASS)
Efficiency Waterfall	src/diagnostics.py	sql/03_metrics.sql	reports/campaign_diagnostic.md	tests/test_diagnostics.py (PASS)
4-Stage Funnel Drop-off	src/diagnostics.py	sql/06_funnel.sql	exports/funnel_analysis.csv	tests/test_funnel.py (PASS)
Direct Standardization	src/diagnostics.py	sql/05_mix_analysis.sql	exports/mix_analysis.csv	tests/test_diagnostics.py (PASS)
Segment Stability (10/25/50)	src/diagnostics.py	sql/04_audience.sql	exports/segment_stability.csv	tests/test_diagnostics.py (PASS)
A/B Testing & Sizing Engine	src/statistics.py	N/A (Analytical)	FastAPI REST Endpoint	tests/test_statistics.py (PASS)
11-Rule Quality Audit	src/validation.py	sql/07_quality.sql	exports/data_quality.csv	tests/test_quality.py (PASS)
FastAPI REST Endpoints	backend/main.py	N/A (Service)	OpenAPI Swagger /docs	tests/test_backend.py (PASS)